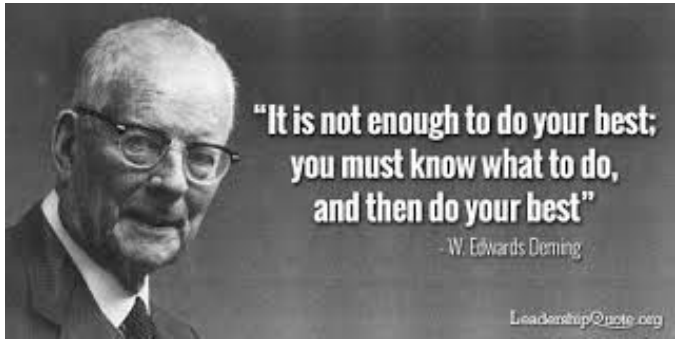
Averaging Data

Calculating the Average (in a data set) is useful because it summarises a large amount of data into a single value, and indicates that variations to this value exists.

Averaging is NOT so useful however, when it comes to measuring performance since it is more probable that results will be ABOVE or BELOW the average. **(you automatically pass or fail!)**

Standard Deviation is more relevant for business prediction and performance review since it calculates a 'likely range'.

William Edwards Deming



Theory of Volatility
…"Volatility is expected and carries with it a degree of predictability; understanding the nature of your volatility is key…"

1. Common Cause: High Frequency – Low Impact . (Ex, Retail store - rainy day)
2. Special Cause: Low Frequency - High Impact. (Ex, Earthquake or Pandemic)
3. Seasonal: Predictable Patterns & Trends (Ex, Selling ice-cream in summer vs. winter)
4. **Tampering:** Data Misinterpretation & Response **(Wrong conclusion = wrong actions)**

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Volatility Options

- Review key products and services - what could help ‘fill in the holes’?
- Review admin procedures - how could we get the money in quicker?
- Review marketing & sales tactics - how could we attract more customers, more often?
- Review Working Capital - how should we finance and manage the money to cope with the ‘swings’?

This presentation focuses on understanding and managing volatility within a business context to improve decision-making. While standard averages provide a simple summary of data, the slides argue that they are insufficient for performance reviews because they fail to account for common fluctuations. Instead, the material promotes the use of standard deviation to establish a likely range of outcomes, offering a more realistic perspective on sales trends.

By analyzing these statistical swings, leaders can identify whether specific results are caused by seasonal patterns or unique events. The final objective is to develop a strategic action plan that addresses marketing, administration, and working capital to better handle future instability.

Summary of the Theory

The core of Deming's Theory of Volatility is about achieving a contextual understanding of business data to avoid misinterpretation.

Many businesses review their performance by plotting sales or revenue data and drawing an average line through it. While an average is useful for summarizing a past dataset, it is completely ineffective for looking forward because it creates an artificial standard where results are automatically judged as a "pass" (above average) or a "fail" (below average).

Instead, the theory advocates for using a Standard Deviation or Volatility Graph, which calculates a "likely range" of performance (e.g., where 68% of results are expected to fall). This helps leaders realize that a business naturally goes up and down, and allows them to maintain an open mind about what is actually causing these variations.

Key Concepts in the Theory:

- Types of Variation: Business volatility is driven by different factors. ***Common variations*** are normal fluctuations (like a rainy day impacting retail). ***Seasonal variations*** are predictable trends. ***Special causes*** are rare, High-impact events like an earthquake or a pandemic.

- **Tampering (Data Misinterpretation):** This is the danger of drawing the wrong conclusion from data and taking the wrong action. For example, a business might see a massive spike in June sales and pay their team a bonus, then see a drop in December and fire staff, completely missing the fact that these swings were just a predictable, seasonal pattern.

- **Causation vs. Correlation vs. Coincidence:** To interpret data properly, you must identify relationships correctly. Causation is a direct link (putting petrol in a diesel engine blows it up). Correlation is a highly attributable contribution (driving with low tire pressure and hitting a pothole causes a flat). Coincidence is unrelated (listening to AC/DC when your engine blows up).

- **The Caravan Metaphor:** Volatility in a business is like a caravan swaying behind a car; if you do not get the caravan's swinging under control, it will pull the entire car off the road. You cannot safely scale or grow a business if you do not understand and manage its underlying volatility first.

Step-by-Step Practical Application

To apply this framework to a business, follow these steps to move from raw data to strategic action.

Step 1: Map Your Data with a Volatility Graph Stop relying solely on averages. Plot your key metrics (like cash received or sales) and add standard deviation lines to establish your upper and lower "likely ranges". This immediately contextualizes your data, showing you what is a normal swing versus what is an anomaly.

Step 2: Analyze the Data Contextually Review the data points regularly and ask critical questions:

- *Is this information consistent with our previous assumptions?*
- *Do any specific data points relate to a special or seasonal cause?*
- *Are our swings influenced by internal campaigns?* (For example, a furniture store running a "free Christmas ham" promotion might see a massive spike in December, but fail to realize it is cannibalizing their regular sales in November and January).

Step 3: Diagnose the Root Cause When you spot anomalies outside your standard deviation, use the framework of Causation, Correlation, and Coincidence to figure out *why* it happened. Ensure you are not "tampering" by jumping to conclusions and punishing or rewarding staff for events out of their control.

Step 4: Apply the Four Volatility Options Once you understand your volatility, apply one of the four key strategies to get the "caravan" back under control:

1. Review Products & Services: Can we introduce new offerings to "fill in the holes" during our slow periods?
2. Review Admin Procedures: Are cashflow swings actually just a result of poor invoicing or slow debt collection? Can we get our money in quicker?
3. Review Marketing & Sales: Do we just need more scale and customer volume to smooth out the bumps?
4. Review Working Capital: If the industry is naturally volatile, how can we better manage retained earnings and overdraft facilities to cope with unavoidable swings?

Common Q&A

Q: Why shouldn't we just measure our success against our monthly average?
A: Averages are useful for summarizing past data, but they fail as a performance metric because they create an automatic "pass or fail" scenario. Since it is highly probable that results will naturally fluctuate above or below the average, using it as a strict baseline leads to poor future forecasting and misinterpreted performance.

Q: What is the risk if we ignore volatility and just focus on growth?
A: Ignoring volatility is like driving a car with a wildly swaying caravan attached; if you try to speed up (scale the business) without fixing the balance, the volatility will pull the business off the road. Extreme swings in sales and cash flow create an incredibly fatiguing and loathsome environment for the business.

Q: How do we know if a drop in sales is a real problem or just a normal fluctuation?
A: You determine this by using a standard deviation graph. This graph establishes a range where 68% of your results are expected to fall based on historical data. If the drop falls within this range, it may just be common variation. If it falls significantly outside the range, you can investigate whether it is due to a seasonal trend, a special cause (like a major market shift), or internal errors.

Q: What is "Tampering"?
A: Tampering occurs when business leaders misinterpret data and take inappropriate action. An example is giving out bonuses because sales spiked in June, and then firing staff because sales dropped in December, when in reality, the business simply experiences a predictable seasonal swing.